Are Self-Evolving Agent Skills Learning Skills?

Anonymous submission

Abstract

We introduce a lifecycle audit for determining whether score gains from self-evolving agent skills represent retained skill improvement. A post-revision score alone is insufficient because optimizer variation, validation selection, execution randomness, and verifier choice can improve the endpoint without creating a durable artifact effect. We combine repeated evaluations, controlled task variations, and lifecycle interventions to examine whether improvements from skill evolution persist and transfer, and to trace their origins to feedback, revision, selection, skill content, and evaluation. The audit separates five evidential claims: selected gain, retained artifact effect, bounded transfer, evolution-pathway dependence, and measurement dependence. This claim-grounded framework lets self-evolving systems report what their experiments establish without equating a better benchmark score with reusable skill learning.

Introduction

Persistent agent skills offer an appealing route to continual improvement. Instead of updating model parameters, an agent can revise an external artifact that stores reusable instructions, scripts, references, templates, or other supporting resources. Recent systems generate, repair, select, and retain such artifacts from execution traces and evaluator feedback, often reporting higher post-evolution benchmark scores as evidence that the agent has learned a better skill (Li and Hu 2026; Liu et al. 2026b; Yang et al. 2026; Xia et al. 2026).

A higher score is not, by itself, an identified skill effect. The selected run may be unusually favorable; the optimizer may produce variable candidates; a validation gate may retain a narrow patch; repeated execution may eventually succeed without a durable behavioral change; and a verifier may reward only part of the intended behavior. These possibilities do not imply that self-evolution is ineffective. They imply that an endpoint score alone cannot locate where the improvement resides.

We therefore ask two connected questions. When self-evolution raises the benchmark score, does the revised skill itself retain a behavioral improvement under fresh deployment and controlled task variation? Regardless of whether it does, which parts of the evolution-and-evaluation lifecycle produced the observed gain?

We organize the experiments around utility, behavioral scope, and lifecycle attribution. Repeated standard-task evaluations of SkillEvolver, SPARK, and CoEvoSkills measure skill utility, and controlled task variants examine how far that utility extends. Inspection of skill content and execution traces then identifies reusable decisions expressed during deployment. Complementary mechanism studies in Skill-Revise, SkillHone, and SkillOpt examine retention and the origins of observed gains. These studies represent trace-conditioned revision, history-based iterative honing, and validation-gated bounded editing in frozen-agent systems with persistent external skill artifacts.

Our audit follows the lifecycle from feedback to revision, selection or adoption, frozen artifact deployment, and measurement. We separate historical success from fresh retention, final-skill utility from actual parent-to-final revision effects, and same-task retention from controlled transfer. We then intervene on feedback, optimizer sampling, gate provenance, artifact components, and verifier choice. A short information-integrity check bounds direct answer or oracle exposure as an alternative explanation; it is not the paper's organizing hypothesis.

The resulting picture is deliberately more qualified than either *self-evolution works* or *self-evolution only exploits the evaluator*. Revised artifacts can cause real behavioral improvements, but the demonstrated effects are often narrow. Single-run or selected gains need not survive fresh deployment. Useful traces and feedback must be converted into an effective edit, adopted into the persistent artifact, and retained behaviorally. Finally, a better score can reflect compatibility with a measurement surface rather than broader task behavior.

Our central contribution is a claim-grounded lifecycle audit: each reported gain is matched to the intervention required to support it and to the strongest conclusion that intervention permits. We make three contributions:

- **Identification framework.** We separate a reported gain from retention, behavioral scope, evolution-and-selection pathway, and measurement dependence.
- **Lifecycle audit.** We apply matched interventions across

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自进化代理的技能是否在学习技能?

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摘要

我们介绍了一种生命周期审计方法，用于判断自进化代理技能带来的分数提升是否代表持久的技能改进。仅凭修订后的分数是不够的，因为优化器变化、验证选择、执行随机性和验证器选择都可能在不创建持久性效果的情况下提高终点分数。我们结合重复评估、受控任务变化和生命周期干预，来检验技能进化带来的改进是否持久且可迁移，并追溯到反馈、修订、选择、技能内容和评估的来源。该审计将证据声明分为五类：选择增益、持久性效果、有界迁移、进化路径依赖和测量依赖。这个基于声明的框架允许自进化系统报告其实验结果，而不将更好的基准分数等同于可重用的技能学习。

引言

持久的代理技能为持续改进提供了一条有吸引力的途径。代理无需更新模型参数，而是可以修订存储可重用指令、脚本、参考资料、模板或其他支持资源的的外部工件。最近的系统从执行轨迹和评估器反馈中生成、修复、选择和保留此类工件，通常报告更高的进化后基准分数，以此作为代理学习了更好技能的证据（Li和Hu 2026；Liu等人 2026b；Yang等人 2026；Xia等人 2026）。

更高的分数本身并不能确定是某种技能效果。选定的运行可能特别有利；优化器可能产生可变候选；验证门可能保留一个狭窄的补丁；重复执行最终可能在没有持久行为变化的情况下成功；并且验证器可能只奖励预期行为的一部分。这些可能性并不意味着自我进化是无效的。它们意味着仅凭终点分数无法定位改进所在的位置。

因此，我们提出两个相关的问题。当自我进化提高基准分数时，修订的技能

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它本身在新的部署和控制任务变化下能否保持行为改进？无论它是否能够做到，进化与评估生命周期中的哪些部分产生了观察到的收益？

我们围绕效用、行为范围和生命周期归因来组织实验。SkillEvolver、SPARK和CoEvoSkills的重复标准任务评估测量技能效用，而受控任务变体则检查这种效用能延伸多远。检查技能内容和执行轨迹可以识别在部署过程中表达的可用决策。Skill-Revise、SkillHone和SkillOpt中的补充机制研究考察了保留情况以及观察到的收益来源。这些研究代表了在具有持久外部技能实物的冻结代理系统中进行的轨迹条件修订、基于历史的迭代精炼和验证门限定的有界编辑。

我们的审计遵循从反馈到修订、选择或采纳、冻结工件部署以及测量的生命周期。我们将历史成功与新鲜留存、最终技能效用与实际父级到最终修订效果、相同任务留存与受控迁移区分开来。然后我们对反馈、优化器采样、门控溯源、工件组件和验证器选择进行干预。一个信息完整性检查将直接答案或或然器暴露作为替代解释进行边界；它不是论文的组织假设。

最终的结果故意比 自我进化有效 或 自我进化仅利用评估者 更有条件。修订的工件可以导致真正的行为改进，但展示的效果往往是狭隘的。单次运行或选定的收益不必在新部署中存活。有用的轨迹和反馈必须转化为有效的编辑、采纳到持久的工件中，并行为上保留。最后，更好的分数可以反映与测量表面的兼容性，而不是更广泛的任务行为。

我们的核心贡献是一个基于主张的生命周期审计：每个报告的收益都与支持它的干预措施以及干预措施允许的最强结论相匹配。我们做出了三项贡献：

- **识别框架**。 我们将报告的增益与保持、行为范围、进化与选择路径以及测量依赖性分开。

• **生命周期审计**。我们针对三种具有代表性的技能进化机制应用匹配的干预措施，而不会将异构研究变成方法排行榜。

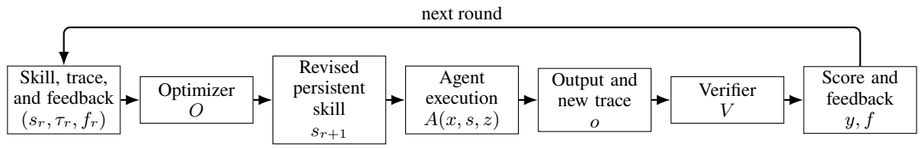


Figure 1: Minimal self-evolution lifecycle. The update operator includes any method-native validation, selection, merge, or adoption before the revised skill is retained.

three representative skill-evolution mechanisms without turning heterogeneous studies into a method leaderboard.

- **Reporting protocol.** We specify the lineage, contrasts, and artifacts required to support each claim level, from selected gain to bounded transfer.

Problem Formulation

Skill Revision and Evaluation

At revision round r, an optimizer updates the current persistent skill artifact:

$$s_{r+1} = O(s_r, \tau_r, f_r), \tag{1}$$

where s_r is the current skill, τ_r is an execution trace, and f_r is feedback that may include a verifier result, a gold answer, a diagnosis, or evaluator feedback. The operator O denotes the method's complete update pathway, including its native merge, validation, selection, or adoption mechanism. A generated candidate that is rejected or never adopted is not s_{r+1}.

The agent then executes task x with the frozen skill:

$$o_r = A(x, s_r, z_r), \tag{2}$$

where z_r represents model sampling, tool use, and environment state, and o_r is the resulting output artifact or state change. A verifier maps the output to a score:

$$y_r = V(x, o_r). \tag{3}$$

The Identification Question

Most evaluations observe:

$$y_{r+1} > y_r. \tag{4}$$

This inequality describes two evaluations; it does not identify what the persistent skill learned. The revised artifact may encode a retained behavioral improvement, the selected score may depend on optimizer or execution variation, or the output may better match the verifier's measurement surface. Infrastructure-invalid observations, such as a verifier that never reaches its semantic assertions, are neither successes nor failure mechanisms and must be repaired or excluded.

Accordingly, we separate two questions:

1. **Effect existence and scope:** Does the revised skill retain a behavioral improvement under fresh deployment, and how far does that effect extend under controlled task variation?
2. **Effect attribution:** Which feedback, revision, selection, artifact, execution, and measurement mechanisms produced the observed score change?

The following experiments address these questions in turn.

Experiments on SkillsBench

Experimental Setup

We evaluate SkillEvolver (Zhang et al. 2026a), SPARK (Zhou et al. 2026b), and CoEvoSkills (Zhang et al. 2026b) on SkillsBench (Li et al. 2026), with two runs per method. Each run pairs an execution using the generated or evolved skill with a no-skill execution of the same task. Both runs of each method use the same task set: 81 tasks for SkillEvolver, 84 for SPARK, and 85 for CoEvoSkills.

We report mean verifier reward on a 0–100 scale and the mean within-task difference between the skill and no-skill conditions. For a run with N tasks, the paired gain is

$$\Delta = \frac{100}{N} \sum_{i=1}^{N} \left(r_i^{\text{skill}} - r_i^{\text{no-skill}} \right). \tag{5}$$

We also count tasks with positive, negative, and zero differences.

Results across Two Runs

All three methods achieve positive mean paired gains in both runs (Table 1). SkillEvolver improves from 30.20 to 55.81 in run 1 and from 37.32 to 62.34 in run 2, yielding gains of 25.61 and 25.02 percentage points. The corresponding task counts are 27 improvements versus six regressions, and 26 improvements versus three regressions.

SPARK increases mean reward from 17.82 to 19.66 in run 1 and from 14.37 to 20.81 in run 2. Its paired gain rises from 1.84 to 6.45 points, with nine versus 13 improved tasks and eight versus five regressions. CoEvoSkills improves from 42.21 to 52.44 and from 23.54 to 33.31 on the aligned 85-task set. Its paired gains are 10.24 and 9.76 points, with 17 improved tasks in each run and nine versus ten regressions.

The direction of the average gain repeats for every method. SkillEvolver's gain remains near 25 points and CoEvoSkills' near ten, while SPARK shows a larger change in gain between runs. Improvements and regressions coexist within each run, making the paired task outcomes informative alongside the aggregate score. The benchmark results quantify the utility of generated and evolved skills under paired deployment. The variant experiments below examine how these gains extend to changed tasks and which artifact-level decisions support transfer.

Generalization under Task Variations

We evaluate source-task skills under changes to input representation (P1), task-instance structure (P2), and coupled task

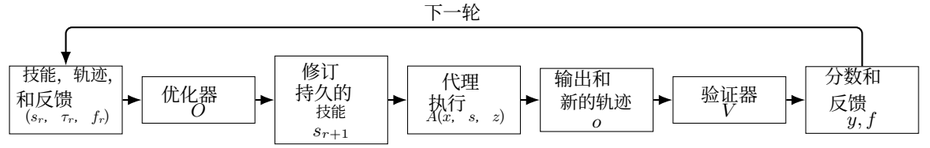


图1：最小自我进化生命周期。更新算子包括修订技能被保留之前的任何方法原生验证、选择、合并或采用。

我们针对三种具有代表性的技能进化机制应用匹配的干预措施，而不会将异构研究变成方法排行榜。

- **报告协议**。我们指定支持从选定增益到有界迁移的每个主张级别的谱系、对比和工件。

问题公式化

技能修订与评估

在修订轮次 r 中，一个优化器更新当前持久的技能工件：

$$s_{r+1} = O(s_r, \tau_r, f_r), \tag{1}$$

其中 s_r 是当前技能，τ_r 是一个执行轨迹，而 f_r 是可能包含验证器结果、标准答案、诊断或评估器反馈的反馈。运算符 O 表示方法完整的更新路径，包括其原生合并、验证、选择或采用机制。一个被拒绝或从未被采用的生成候选不是 s_{r+1}。

代理随后使用冻结的技能执行任务 x：

$$o_r = A(x, s_r, z_r), \tag{2}$$

其中 z_r 表示模型采样、工具使用和环境状态，而 o_r 是结果输出工件或状态变化。一个验证器将输出映射为一个分数：

$$y_r = V(x, o_r). \tag{3}$$

识别问题

大多数评估观察到：

$$y_{r+1} > y_r. \tag{4}$$

这个不等式描述了两种评估；它并不能识别持久技能学到了什么。修订后的工件可能编码了保留的行为改进，选定的分数可能取决于优化器或执行变化，或者输出可能更好地匹配验证器的测量表面。基础设施无效的观察，例如从未达到其语义断言的验证器，既不是成功机制也不是失败机制，必须被修复或排除。

因此，我们区分了两个问题：

1. **效果存在和范围：** 修订后的技能在全新部署下是否保留了行为改进，以及这种效果在受控任务变化下会延伸多远？2. **效果归因：** 哪些反馈、修订、选择、工件、执行和测量机制导致了观察到的分数变化？

以下实验逐一回答这些问题。

SkillsBench上的实验

实验设置

我们在SkillsBench（Li等人，2026年）上评估SkillEvolver（张等人，2026a）、SPARK（周等人，2026b）和CoEvoSkills（张等人，2026b），每种方法运行两次。每次运行都将使用生成的或进化的技能执行与相同任务的技能执行配对。每种方法的两次运行都使用相同的任务集：SkillEvolver有81个任务，SPARK有84个任务，CoEvoSkills有85个任务。

我们报告了验证器在0-100分制上的平均奖励，以及技能条件和无技能条件之间的平均任务差异。对于包含 N 个任务的运行，配对增益为

$$\Delta = \frac{100}{N} \sum_{i=1}^{N} \left(r_i^{\text{skill}} - r_i^{\text{no-skill}} \right). \tag{5}$$

我们还统计了具有正、负和零差异的任务数量。

两次运行的结果

所有三种方法在两次运行中都实现了正的平均配对增益（表1）。SkillEvolver在运行1中从30.20改进到55.81，在运行2中从37.32改进到62.34，分别获得了25.61和25.02个百分点增益。相应的任务计数为27次改进与6次退化，以及26次改进与3次退化。

SPARK在运行1中将平均奖励从17.82提高到19.66，在运行2中从14.37提高到20.81。其配对增益从1.84提高到6.45分，有9个任务改进与13个任务退化，以及8个任务改进与5个任务退化。CoEvoSkills在对齐的85任务集上从42.21提高到52.44，从23.54提高到33.31。其配对增益为10.24和9.76分，每次运行有17个任务改进，以及9次改进与10次退化。

每种方法的平均增益方向都重复出现。SkillEvolver的增益保持在25分左右，而CoEvoSkills的增益更接近，SPARK则在运行之间显示出更大的增益变化。每次运行中，改进和退化并存，使得配对任务结果与汇总分数同样具有参考价值。基准测试结果量化了在配对部署下生成的和进化的技能的效用。下方的变体实验将检验这些增益如何扩展到变化的任务，以及哪些组件级别的决策支持迁移。

任务变化下的泛化能力

我们评估在输入表示（P1）、任务实例结构（P2）和耦合任务变化下的源任务技能

Method	Run	Tasks	No skill	Skill	Δ	Improved	Regressed	Tied
SkillEvolver	1	81	30.20	55.81	+25.61	27	6	48
SkillEvolver	2	81	37.32	62.34	+25.02	26	3	52
SPARK	1	84	17.82	19.66	+1.84	9	8	67
SPARK	2	84	14.37	20.81	+6.45	13	5	66
CoEvoSkills	1	85	42.21	52.44	+10.24	17	9	59
CoEvoSkills	2	85	23.54	33.31	+9.76	17	10	58

Table 1: Two runs per method on SkillsBench. Scores are mean verifier rewards multiplied by 100; Δ is the mean paired gain in percentage points. Each method uses an identical task set across its two runs.

Method	Variants	No skill	Skill	Δ	Improved	Regressed	Tied
SkillEvolver	222	41.87	53.81	+11.94	47	20	155
SPARK	221	41.61	42.25	+0.65	32	37	152
CoEvoSkills	221	41.61	36.60	−5.01	26	41	154

Table 2: Descriptive summary of historical and recovery evaluations on task variants. Each method is paired with its own same-variant no-skill baseline. Scores and within-variant differences use the scale in Table 1.

requirements (P3). The final record covers 222 task variants and 886 condition-level results across no skill and the three methods, including completed recovery evaluations. For each method, Table 2 summarizes the recorded reward difference from a no-skill execution of the same variant.

SkillEvolver. Across 222 variants, mean recorded reward increases from 41.87 to 53.81, a mean within-variant gain of 11.94 points, with 47 improvements and 20 regressions. Gains remain positive under all three types of change: 11.85 points on P1, 14.56 on P2, and 9.72 on P3. The strongest gain occurs when the task instance changes, consistent with learned procedures adapting beyond the original inputs.

SPARK. Across 221 variants, mean recorded reward increases from 41.61 to 42.25, a mean within-variant gain of 0.65 points, with 32 improvements and 37 regressions. Gains are 2.00 points on P1 and 4.53 on P3, while P2 produces a 5.23-point decline.

CoEvoSkills. Across 221 variants, mean recorded reward decreases from 41.61 to 36.60, a mean within-variant change of -5.01 points, with 26 improvements and 41 regressions. Changes are -3.59 points on P1, -9.51 on P2, and -2.42 on P3. The positive gains on standard tasks do not extend to the changed tasks in this summary.

Skill Reuse under Task Variation

SkillEvolver's positive gains across P1–P3 indicate that the utility of its frozen skills extends beyond the source instances. The weaker or negative variant gains of SPARK and CoEvoSkills show that improvement on the original task and usefulness after task changes are distinct properties. This makes the content preserved during reuse central to interpreting the results.

Two complementary forms of content emerge from the inspected skill files and execution traces: diagnostic rules for choosing and revising a procedure, and constraints for checking the resulting output. In the examined SkillEvolver cases, these rules remain explicit while input handling and solution construction adapt to the variant. Corresponding failure traces reveal that reusing a procedure can preserve much of the workflow while missing a decision or constraint needed for correctness. These observations suggest that reusable decision rules, including when they apply and how their outputs should be checked, are a useful unit of analysis for skill transfer.

Within the lifecycle audit, variant performance characterizes the scope of observed utility. The subsequent sections, *Identifying Retained Skill Effects* and *Attributing Observed Skill Gains*, examine persistence under fresh deployment and the contributions of feedback, revision, selection, skill content, and evaluation.

Identifying Retained Skill Effects

Evaluation Design and Contrasts

We analyze three persistent-skill systems with distinct revision mechanisms. SkillRevise converts execution traces and diagnoses into bounded repair principles (Liu et al. 2026b). SkillHone accumulates practice history and iteratively reflects on the persistent skill (Li and Hu 2026). SkillOpt produces structured bounded edits and uses a validation gate and rejected buffer before adoption (Yang et al. 2026). The systems are studied as representative mechanisms, not as directly comparable leaderboard entries.

At the paper level, the evidence is exploratory: individual interventions use frozen inputs and predeclared gates, but the combined studies are designed to identify candidate mechanisms rather than estimate benchmark-wide prevalence. A targeted 24-task SkillsBench panel tests whether historical outcomes survive fresh deployment. The panel is stratified to expose stability failures and method disagreements; it is not a random sample. Three controlled mechanism cases then compare complete parent and final artifacts under predeclared variations. Within each contrast, we hold the executor model, harness, task instance, budget, and verifier version

方法	Run	任务	无技能	技能	Δ	改进	退化	Tied
SkillEvolver	1	81	30.20	55.81	+25.61	27	6	48
SkillEvolver	2	81	37.32	62.34	+25.02	26	3	52
SPARK	1	84	17.82	19.66	+1.84	9	8	67
SPARK	2	84	14.37	20.81	+6.45	13	5	66
协同进化技能	1	85	42.21	52.44	+10.24	17	9	59
协同进化技能	2	85	23.54	33.31	+9.76	17	10	58

表1：SkillsBench上每个方法运行两次。分数是验证器奖励的平均值乘以100； Δ 是百分比点上的平均配对增益。每种方法在其两次运行中使用相同的任务集。

方法	变体	无技能	技能	Δ	改进	退化	Tied
SkillEvolver	222	41.87	53.81	+11.94	47	20	155
SPARK	221	41.61	42.25	+0.65	32	37	152
协同进化技能	221	41.61	36.60	−5.01	26	41	154

表 2：任务变体历史和恢复评估的描述性摘要。每种方法都与其自身的同变体无技能基线配对。分数和变体内差异使用表 1 中的量表。

最终记录涵盖了222种任务变体和886种条件级结果，包括无技能和三种方法（包括完成的恢复评估）。对于每种方法，表2总结了与相同变体的无技能执行相比记录的奖励差异。

SkillEvolver. 在 222 个变体中，平均记录的奖励从 41.87 增加到 53.81，平均变体内收益为 11.94 分，有 47 次改进和 20 次退化。在所有三种类型的变更下，收益保持正值：P1 上为 11.85 分，P2 上为 14.56 分，P3 上为 9.72 分。最大的收益发生在任务实例变更时，这与学习到的程序适应并超越了原始输入相一致。

SPARK. 在221种变体中，平均记录的奖励从41.61增加到42.25，平均变体内收益为0.65分，其中32次改进和37次退化。在P1上收益为2.00分，在P3上为4.53分，而P2产生了5.23分的下降。

协同进化技能。 在221个变体中，平均记录的奖励从41.61降至36.60，平均变体内变化为 -5.01 点，其中26次改进和41次退化。变化在P1上是 -3.59 点，在P2上是 -9.51 ，在P3上是 -2.42。在标准任务上的正面收益并未扩展到本摘要中的变更任务。

任务变化下的技能重用

SkillEvolver在P1–P3上的正向增益表明其冻结技能的效用超出了源实例。SPARK和Co- EvoSkills较弱的或负向的变体增益表明，在原始任务上的改进和在任务变化后的实用性是不同的属性。这使得在重用时保留的内容成为解释结果的关键。

从检查的技能文件和执行轨迹中，出现了两种互补的内容形式：用于选择和修订程序的诊断规则，以及用于检查最终输出的约束。在考察的SkillEvolver案例中，这些规则保持明确，而输入处理和解决方案构建会适应变体。相应的失败轨迹显示，重用程序可以保留大部分工作流程，但缺少正确性所需的决策或约束。这些观察表明，可重用的决策规则（包括它们何时适用以及如何检查其输出）是技能迁移的有用分析单元。

在生命周期审计中，变体性能体现了观察到的效用范围。后续章节《识别保留的技能效果》和《归因观察到的技能增益》将探讨全新部署下的持久性，以及反馈、修订、选择、技能内容和评估的贡献。

识别保留的技能效果

评估设计与对比

我们分析了三种具有不同修订机制的持久技能系统。SkillRevise将执行轨迹和诊断转化为有界修复原则（Liu等人，2026b）。SkillHone积累练习历史并迭代地反思持久技能（Li和Hu，2026）。SkillOpt生成结构化有界编辑，并在采纳前使用验证门和拒绝缓冲区（Yang等人，2026）。这些系统作为代表性机制被研究，而不是作为可直接比较的排行榜条目。

在论文层面，证据具有探索性：单个干预使用冻结的输入和预声明的门，但综合研究旨在识别候选机制而非估计基准范围内的普遍性。一个有针对性的24任务SkillsBench面板测试历史结果是否能在全新部署中存活。该面板分层以暴露稳定性失效和方法分歧；它不是一个随机样本。随后三个受控机制案例比较在预声明变化下完整的父级和最终工件。在每个对比中，我们保持执行者模型、适配器、任务实例、预算和验证器版本不变，除非其中一个是命名干预；确切的提示、哈希值、种子和停止决策都保留在工件包中。任务是主要分析单元，而重复运行是相同任务-条件对的部署。

Panel and contrast	Main result	Supported inference and boundary
Panel A: Retention under fresh deployment		
Historical pass → fresh repeats	12/24 tasks flip within at least one frozen condition; selected-outcome retention is 47.1% for SkillRevise and 55.0% for SkillHone	A selected pass is not a deployment probability; denominators differ and are not a method ranking.
SkillRevise final → no-skill	16/48 vs. 14/48; +4.2 pp, 95% CI $[-10.4, 20.8]$	No clear fresh-utility advantage on this targeted panel; not a parent-to-final revision estimate.
SkillHone final → no-skill	12/48 vs. 14/48; −4.2 pp, 95% CI $[-20.8, 14.6]$	No clear fresh-utility advantage on this targeted panel; not a parent-to-final revision estimate.
Panel B: Revision effects under controlled variations		
Invoice parent → final	$0/12 \rightarrow 12/12$; three new same-contract corpora: $0/9 \rightarrow 9/9$	Stable but narrow same-contract transfer.
Energy parent → final	Original task plus 5% parameter variant: $1/6 \rightarrow 5/6$	Directional parameter transfer in a small stochastic case.
SkillOpt SearchQA parent → final	Independent held-out $9/24 \rightarrow 17/24$; stable-transfer gate fails	Local held-out gain, not stable general procedure learning.

Table 3: Evidence for retained skill effects. Panel A contains 24 targeted tasks, three conditions, and two fresh deployments per condition (144/144 behavior-valid runs). Different historical-retention denominators preclude horizontal method ranking. The no-skill condition calibrates execution and selection variability; final-versus-no-skill does not identify revision effects.

fixed unless one is the named intervention; exact prompts, hashes, seeds, and stopping decisions are preserved in the artifact package. The task is the primary analysis unit, while repeated runs are deployments of the same task-condition pair.

Four comparisons must remain distinct. Historical-pass to fresh-repeat comparisons test retention of a selected outcome, not advantage over a baseline. Final-to-no-skill comparisons estimate fresh utility, not the effect of revision. Only actual-parent to final comparisons identify a revision effect. Applying that comparison to controlled variants tests bounded transfer, not population-average benefit.

Retention under Fresh Deployment

This experiment tests whether historically selected outcomes survive when the final artifact is frozen and redeployed. Holding task, skill, and condition fixed does not make execution deterministic. Across the 24-task panel, 12 tasks change success status between their two repeats in at least one condition. Aggregate fresh success is 14/48 without a skill, 16/48 with the SkillRevise final artifact, and 12/48 with the SkillHone final artifact. Both matched-lift intervals cross zero (Table 3).

These results establish an identification failure in the common reporting practice: cached or selected success cannot be read as stable deployment capability. They do not establish that either revision method is ineffective. The panel is purposive, and it lacks each method's actual parent condition. Its proper conclusion is narrower: on this sample, neither final-versus-no-skill comparison establishes a clear average fresh-deployment advantage, and fresh repeats are necessary before a selected score is called retained.

Revision Effects under Controlled Variations

These mechanism probes test whether the actual parent-to-final revision effect persists under predeclared task changes. In Invoice, the complete final skill repairs all 12 matched deployments and retains the effect on nine deployments drawn from three disjoint corpora that preserve the contract while changing the documents. This is strong evidence of a stable, same-contract revision effect, but not of broad cross-contract generalization.

The Energy case provides weaker but directionally consistent evidence: the complete final artifact improves the original setting and a 5% parameter variant in five of six deployments, compared with one of six for the parent. The sample is too small to support a prevalence or generality claim. In SkillOpt SearchQA, the final artifact improves independent held-out accuracy from 9/24 to 17/24, yet fails the predeclared stability threshold. It therefore supports local held-out benefit rather than stable transferable procedure learning.

Taken together, revised skill artifacts can produce retained behavioral effects, but a selected benchmark pass alone does not identify such an effect, and the effects established so far remain narrow rather than broadly generalizable.

Attributing Observed Skill Gains

Lifecycle Interventions

The experiments in this section intervene at different lifecycle stages. They do not estimate one common effect size. Each intervention changes one object while holding its relevant lineage fixed: feedback-view studies hold the parent and failures fixed; optimizer-repeat studies hold the input evidence fixed; gate audits preserve accepted and rejected candidates;

面板与对比	主要结果	支持推理与边界
面板A：全新部署下的留存率		
历史通过 → 全新重复	12/24任务至少翻转一种冻结条件；选择结果保留是SkillRevise为47.1%，SkillHone为55.0%	一个选择通过不是部署概率；分母不同，也不是方法排名。
SkillRevise 最终 → 无技能	16/48 vs. 14/48；+4.2pp, 95% CI $[-10.4, 20.8]$	在此目标面板上没有明显的全新实用优势；不是从父级到最终修订的估计。
SkillHone 最终 →无技能	12/48 与 14/48； −4.2 pp，95% CI $[-20.8, 14.6]$	在该目标面板上，没有明显的全新效用优势；不是从父版到最终修订的估计。
面板 B：受控变化下的修订效果		
发票父版 →最终	$0/12 \rightarrow 12/12$; 三个新的相同合同语料库：$0/9 \rightarrow 9/9$	稳定但狭窄的相同合同迁移。
能量父 →最终	原始任务加5%参数变体: $1/6 \rightarrow 5/6$	小随机情况下的方向参数迁移。
SkillOpt SearchQA 父 → 最终	独立保留集 $9/24 \rightarrow 17/24$; 稳定迁移门控失败	本地保留集增益，非稳定泛化程序学习。

表3：保留技能效应的证据。面板A包含24个目标任务、三种条件和每个条件下的两次全新部署（144/144行为有效运行）。不同的历史保留分母排除了横向方法排序。无技能条件校准执行和选择变异性；最终与无技能的对比无法识别修订效应。

固定，除非其中一个是命名干预；确切的提示、哈希值、种子和停止决策都保留在工件包中。任务是主要分析单元，而重复运行是相同任务-条件对的部署。

必须保持四个比较的独立性。历史通过到全新重复的比较测试选定结果的保留，而非相对于基线的优势。最终到无技能的比较估计全新效用，而非修订效应。只有实际父代到最终的比较能识别修订效应。将该比较应用于受控变体测试有界迁移，而非群体平均收益。

全新部署下的保留

本实验测试历史选定结果在最终产品冻结并重新部署时是否存活。固定任务、技能和条件并不能使执行确定性。在24任务面板中，至少有一种条件下，12个任务在两次重复之间改变了成功状态。无技能的全新成功率是14/48，SkillRevise最终产品是16/48，SkillHone最终产品是12/48。两个匹配提升间隔都穿过零（表3）。

这些结果表明在常规报告实践中存在识别失败：缓存的或选定的成功不能被视为稳定的部署能力。它们并未表明任何修订方法都无效。该小组是故意的，并且缺乏每种方法的真实父条件。其恰当的结论更为狭窄：在这个样本中，无论是最终版本与无技能比较，都无法明确证明平均新部署优势，并且在新重复之前，不能将选定的分数称为保留。

在受控变化下的修订效果

这些机制探测测试在实际父版本到最终修订效果在预先声明的任务变化下是否持续存在。在发票中，完整的最终技能修复了所有12个匹配部署，并保留了从三个保持合同但更改文档的独立语料库中抽取的九个部署的效果。这是稳定、同一合同修订效果的强有力证据，但不是跨合同泛化的证据。

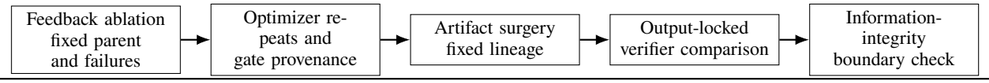
“能量”案例提供了较弱但方向上一致的证据：最终完整成果改进了原始设置，在六次部署中的五次出现了5%的参数变体，而父代案例仅出现一次。样本量太小，无法支持普遍性或一般性主张。在SkillOpt SearchQA中，最终成果将独立保留集准确率从9/24提升至17/24，但未能通过预定义的稳定性阈值。因此它支持局部保留集收益，而非稳定的可迁移程序学习。

综合来看，修订的技能成果可以产生保留的行为效果，但仅通过一个选定的基准测试并不能识别此类效果，且迄今为止建立的效果仍较为狭窄而非广泛可推广。

归因于观察到的技能提升

生命周期干预

本节中的实验在不同的生命周期阶段进行干预。它们不估计一个共同的效应量。每次干预都会改变一个对象，同时保持其相关谱系固定：反馈-视图研究保持父对象和失败案例固定；优化器-重复研究保持输入证据固定；门控审计保留接受的和被拒绝的候选对象；



Lifecycle stage	SkillRevise	SkillHone	SkillOpt
Feedback and revision	Diagnosis ablation; revision traces	Feedback-view ablation	Feedback branches; optimizer repeats
Selection and adoption	Revision history	Iterative honing history	Native validation gate; accepted and rejected candidates
Artifact mediation	Not yet matched at the same strength	Invoice and Energy surgery	SearchQA component surgery
Measurement	No output-locked comparison yet	Radar verifier swap	Spreadsheet dual scoring
Information integrity	Artifact forensic	Artifact forensic	Wrong-metadata and oracle controls

Figure 2: Lifecycle attribution design and current evidence coverage. The upper panel locates each intervention; the lower panel records method coverage. Populated cells do not imply equivalent strength, and missing cells bound cross-method claims.

artifact surgery holds the surrounding artifact fixed; and verifier counterfactuals hold the output fixed.

Figure 2 makes the current coverage and its gaps explicit. The shared audit logic is that feedback access, candidate revision, validation acceptance, adoption into the persistent artifact, and retained behavioral improvement are distinct events.

Revision and Selection Pathways

These interventions test whether the observed endpoint depends on the feedback view, optimizer sample, or selection gate. In SkillRevise, the complete three-revision pipeline succeeds on 53/86 tasks, compared with 28/86 when diagnosis is removed. Task-level discordance is 26 gains versus one loss, and 2,326 of 3,648 revision entries explicitly use verifier evidence. This demonstrates that diagnosis changes the selected endpoint and that the revision process is strongly evaluator-facing. It does not, by itself, establish which changes are retained under fresh deployment.

SkillHone separates access to feedback from stable benefit. Detailed sanitized, binary, and redacted-tail feedback obtain first-deployment success on 4/12, 4/12, and 5/12 cases, while four of 18 repeated cells flip. Feedback can trigger revisions, but these data do not support a stable advantage for one feedback view.

SkillOpt further separates candidate production from attribution and adoption. In matched SearchQA optimizer repeats, raw-gold and outcome-only views each produce four applied edits. Yet the preregistered between-condition effect size is only $D = .044 < .20$, despite weak lexical decodability, so a single patch cannot be cleanly attributed to feedback view. Across outcome-only and trajectory-only branches, all 6/6 candidates change and 3/6 pass the gate, but every acceptance is driven by the same Blackwater validation item. In OfficeQA, a selected one-item gain disappears in fresh matched deployment. Thus edit production, gate acceptance, and retained improvement cannot be collapsed into one event.

Artifact-Level Mediation

This experiment tests whether a feedback-derived component, rather than unrelated edits in the full artifact, causes the behavior after adoption. Let P denote the complete parent, F the complete final artifact, $P+B$ the parent with component B, and $F-B$ the final artifact with that component removed.

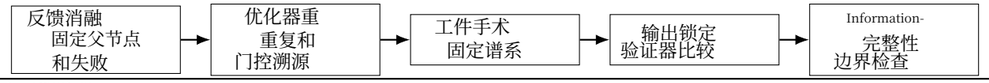
Case	Result	Maximum claim
Invoice	$P = 0/12$, $P+B = 12/12$, $F-B = 0/12$, $F = 12/12$	B is necessary and sufficient for the narrow same-contract effect.
Energy	Local mediation in the original task; necessity and sufficiency do not survive the variant	Local component effect, not transferable mediation.
SearchQA	9 repairs, 0 regressions; stable success 7/12, below the strong gate	Directional mediation on part of the held-out set.

Table 4: Artifact-level mediation evidence.

These interventions establish that an adopted artifact component can cause behavior. They do not show that the same mechanism is prevalent across methods. In particular, Skill-Revise lacks a matched surgery of equal strength, and the current cases should be read as existence and boundary evidence.

Measurement Dependence under Fixed Outputs

This counterfactual tests whether a verdict changes when the output is fixed and only the measurement changes. In the Radar family, all 12 reconstructed outputs reproduce their historical original-verifier outcomes. Of six original passes, only one remains a pass under a reference-calibrated strict verifier; the five valid flips arise from specification-implied



生命周期阶段	SkillRevise	SkillHone	SkillOpt
反馈与修订	诊断消融；修订轨迹	反馈视图消融	反馈分支；优化器重复
选择与采纳	修订历史	迭代打磨历史	本地验证门；已接受和被拒绝的候选者
工件调解	尚未在相同力量	发票和能源手术	SearchQA组件手术
测量	尚未有输出锁定比较	雷达验证器交换	电子表格双重评分
信息完整性	文物取证	文物取证	错误元数据和预言机控制

图2：生命周期归因设计及当前证据覆盖范围。上层面板定位每个干预措施；下层面板记录方法覆盖范围。填充的单元格不表示同等强度，缺失的单元格界定跨方法主张。

工件手术保持周围工件固定；并且验证器反事实研究保持输出固定。

图2使当前的覆盖范围及其空白处变得明确。共享的审计逻辑是：反馈访问、候选修订、验证接受、采用到持久化工件以及保留的行为改进是不同的独立事件。

修订与选择路径

这些干预措施测试观察到的终点是否依赖于反馈视图、优化器样本或选择门。在SkillRevise中，完整的三次修订管道在53/86个任务上成功，而移除诊断时只有28/86个任务成功。任务级别的分歧是26个收益与1个损失，且在3,648个修订条目中，有2,326个明确使用了验证器证据。这表明诊断改变了选定的终点，且修订过程强烈依赖于评估者。它本身并不能确定哪些变更会在全新部署时被保留。

SkillHone将反馈访问与稳定收益分离。详细的清理版、二进制版和截尾红移版反馈在4/12、4/12和5/12的案例中获得首次部署成功，而18个重复单元中有4个发生了翻转。反馈可以触发修订，但这些数据不支持任何一种反馈视图具有稳定优势。

SkillOpt 进一步将候选生成与归因和采纳分开。在匹配的 SearchQA 优化器重复中，原始金标和仅结果视图各自产生四个应用过的编辑。然而，预先注册的组间效应量仅为 $D = .044 < .20$，尽管词汇解码能力较弱，因此单个补丁无法清晰地归因于反馈视图。在仅结果分支和仅轨迹分支中，所有 6/6 候选都发生了变化，3/6 通过了门控，但每次接受都是由同一个 Blackwater 验证项驱动的。在 OfficeQA 中，一个被选中的单项增益在新的匹配部署中消失。因此，编辑生成、门控接受和保留的改进不能合并为一个事件。

组件级调解

这项实验测试了反馈派生的组件，而不是完整工件中的无关编辑，是否导致了采纳后的行为。令 P 表示完整父项，F 表示完整最终工件，$P+B$ 表示包含组件 B 的父项，以及 $F-B$ 表示移除该组件的最终工件。

Case	结果	最大主张
发票	$P = 0/12$, $P+B = 12/12$, $F-B = 0/12$, $F = 12/12$	B 是必要且充分的。对于狭义的同合约效应。
能量	本地调解在原始任务；必要性和充分性无法存活变异体	局部组件效应，不可转移的调解。
SearchQA	9个修复，0回归错误；稳定成功7/12，低于强门	部分方向中介保留集。

表4：工件级中介证据。

这些干预措施表明，一个采用的工件组件可以导致行为。它们并未表明相同的机制在所有方法中都普遍存在。特别是，Skill-Revise缺乏同等强度的匹配手术，当前案例应被视为存在性和边界证据。

固定输出下的测量依赖性

这个反事实测试了当输出固定而仅测量变化时，判决是否会改变。在Radar系列中，所有12个重建输出都重现了其历史原始-验证器结果。在六个原始通过中，只有一个在参考校准的严格验证器下仍然通过；五个有效翻转源于规范隐含的

checks. The same behavior can therefore receive different valid verdicts because the measurement surfaces differ.

Measurement dependence can also operate in the other direction. For identical Spreadsheet workbook outputs, the reported value-based evaluator scores the complete SkillOpt artifact at 10/12, whereas joint value-and-formula recalculation scores it at 8/12. Representation, cache materialization, or formula inspection can therefore change what the evaluator observes. This dual scoring is not identical in design to the Radar output-locked swap, but it reinforces the narrower point that score is a property of output plus measurement, not behavior alone.

Neither result proves semantic failure, exploitation, or benchmark-wide prevalence. To identify evolution-induced verifier dependence, one would additionally need matched no-skill, parent, and final outputs under both original and calibrated verifiers, and test whether the parent-to-final verifier gap increases.

Information-Integrity Boundary

This supporting check tests whether restricted answer or oracle information enters and controls the artifact. High-confidence direct-answer or oracle matches are absent from 86 SkillRevise and 85 SkillHone natural artifacts. SkillOpt conditions containing the feedback-derived answer-surface section follow incorrect metadata in 0/15 probes. In contrast, an unsafe hidden-oracle positive control changes probe success from 0/5 to 5/5 and leaves private-oracle or gold markers.

These checks show that the audit can detect direct exposure when it is deliberately introduced, while the natural artifacts do not support direct copying as a universal explanation. They do not prove the absence of implicit answer-following policies or all forms of leakage.

Observed score gains can arise at multiple lifecycle stages: feedback shapes candidate revisions, selection can amplify narrow improvements, adopted artifact components can mediate real behavior, and valid verifiers can disagree on fixed outputs. These mechanisms must be identified separately before a score gain is interpreted as skill learning.

From Reported Gains to Auditable Claims

Evidence Requirements and Claim Boundaries

The audit is not a scalar ladder from *safe* to *unsafe*. Its steps answer independent questions and impose distinct claim boundaries.

A selected gain is weaker than a retained artifact effect; a retained effect on one instance is weaker than controlled transfer. Evolution-pathway and measurement claims remain meaningful even when retention is not established. Conversely, verifier-aware language or evaluator dependence must not be relabeled as leakage without restricted-information provenance and behavioral evidence.

Lifecycle Reporting Protocol

An auditable self-evolution report should preserve the following minimum evidence:

Audit step	Required test → maximum supported claim
Validity	Execution and verifier status plus assertion coverage → eligible observation
Retention	Frozen actual-parent/final fresh repeats → retained artifact effect
Behavioral scope	Entity, parameter, input, contract, or family variants → bounded controlled transfer
Evolution pathway	Feedback ablation, optimizer repeats, and gate provenance → pathway dependence or selection mediation
Measurement	Output-locked calibrated verifier comparison → measurement dependence
Information integrity	Provenance, wrong-metadata, and positive control → exploitation or confirmed leakage, when supported

Table 5: Evidence requirements and claim boundaries. The rows answer complementary questions rather than forming a scalar risk ladder.

- **Before optimization:** parent artifact, task and split identity, failure pool, environment, and verifier version.
- **During revision:** optimizer-visible traces and feedback, candidate patches, optimizer samples, prompts, and usage.
- **At selection and adoption:** accepted and rejected artifacts, gate items, per-item decisions, and exact adopted artifact.
- **At deployment:** frozen artifact, actual-parent and no-skill conditions, fresh repeats, controlled variants, and output bytes.
- **At scoring:** verifier logs, assertion coverage, reference calibration, and any alternative verifier applied to the same output.
- **For cost:** separate optimizer, executor, verifier, and invalid-attempt ledgers.

The complete artifact appendix should additionally record hashes, commands, seeds, environment revisions, timeouts, and stopping decisions. This protocol turns a post-evolution score into an auditable lineage and limits the paper's language to the strongest matched contrast actually performed.

Related Work

Skill Evolution and Empirical Evaluation

Agent-skill research spans generation, revision, selection, retrieval, organization, and reinforcement learning. SkillsBench and SWE-Skills-Bench compare skill and no-skill utility across diverse or software-engineering tasks (Li et al. 2026; Han et al. 2026). SkillLearnBench evaluates continual skill generation (Zhong et al. 2026), while SkillRevise, SkillHone, SkillOpt, SkillRL, and CoEvoSkills propose different revision or learning loops (Liu et al. 2026b; Li and Hu 2026; Yang et al. 2026; Xia et al. 2026; Zhang et al. 2026b). Recent empirical work broadens the object of study: From Raw Experience to Skill Consumption separates experience generation, extraction, and consumption (Huang et al. 2026); How Well Do Agentic Skills Work in the Wild studies

检查。因此，相同的行为可能会因为测量表面不同而获得不同的有效判决。

测量依赖性也可以反向运作。对于相同的电子表格工作簿输出，基于值的评估器将完整的 SkillOpt 实物评为 10/12，而联合值和公式重新计算则将其评为 8/12。因此，表示、缓存物化或公式检查可以改变评估器所观察到的内容。这种双重评分在设计上与雷达输出锁定交换并不相同，但它加强了更狭窄的观点：分数是输出加上测量的属性，而不仅仅是行为。

Neither结果证明语义失败、利用或基准范围内的普遍存在。为了识别进化引起的验证器依赖，还需要在原始和校准的验证器下匹配无技能、父代和最终输出，并测试父代到最终验证器差距是否增加。

信息完整性边界

此辅助检查测试是否受限答案或预言机信息进入并控制该工件。高置信度的直接答案或预言机匹配在86个SkillRevise和85个SkillHone自然工件中不存在。包含反馈派生答案表面部分的SkillOpt条件在15次探测中有0次遵循错误元数据。相比之下，一个不安全的hidden-oracle正向控制将探测成功率从0/5变为5/5，并留下private-oracle或gold标记。

这些检查表明，审计可以检测到当直接暴露被故意引入时，而自然产生的伪影并不支持直接复制作为普遍解释。它们并不能证明隐式答案跟随策略或所有形式的泄漏不存在。

观察到的分数增益可能出现在多个生命周期阶段：反馈塑造候选修订，选择可以放大狭窄的改进，采用的伪影组件可以调解实际行为，有效的验证器可以就固定输出存在分歧。在将分数增益解释为技能学习之前，必须单独识别这些机制。

从报告的增益到可审计的声明证据要求和声明边界

审计不是从安全到不安全的标量阶梯。它的步骤回答独立的问题，并施加不同的声明边界。

一个选定的增益比保留的伪影效应弱；对一个实例的保留效应比受控迁移弱。即使保留没有建立，进化路径和测量声明仍然是有意义的。相反，验证器感知语言或评估器依赖必须在没有受限制信息的来源和行为证据的情况下，不能被重新标记为泄漏。

生命周期报告协议

一份可审计的自进化报告应保留以下最低证据：

审计步骤	必须测试 → 最大支持的声明
有效性	执行和验证器状态以及断言覆盖率 → 符合资格
留存	观察 冻结实际父节点/最终新鲜重复 → 保留的工件
行为	效果 实体、参数、输入、合约或家族变体 → 限定
范围	受控迁移
进化	反馈消融、优化器重复以及门控溯源 →
路径	路径依赖或选择介导
测量	输出锁定校准验证器比较 →测量
信息完整性	依赖 来源、错误元数据和正向控制 → 利用或确认泄露，当受支持时

表 5：证据要求和声明边界。行回答互补问题，而不是形成一个标量风险阶梯。

- **优化前：** 父级工件、任务和分身份、失败池、环境以及验证器版本。
- **在修订期间：** 优化器可见的轨迹和反馈、候选补丁、优化器样本、提示和使用。
- **在选择和采用期间：** 接受的和拒绝的工件、门控项、逐项决策和精确采用的工件。
- **在部署期间：** 冻结的工件、实际父和技能无条件、新鲜重复、受控变体和输出字节。
- **在评分期间：** 验证器日志、断言覆盖率、参考校准和应用于相同输出的任何替代验证器。
- **关于成本：** 单独的优化器、执行器、验证器和无效尝试账本。

完整的工件附录还应记录哈希值、命令、种子、环境修订、超时和停止决策。该协议将进化后的分数转化为可审计的谱系，并将论文的语言限制为实际执行的最强匹配对比。

相关工作

技能进化和实证评估

代理技能研究涵盖生成、修订、选择、检索、组织以及强化学习。技能基准（Skill Bench）和软件工程技能基准（SWE-Skills Bench）比较技能与无技能在不同或软件工程任务中的效用（Li 等人，2026；Han 等人，2026）。SkillLearnBench评估持续技能生成（Zhong 等人，2026），而SkillRevise、SkillHone、SkillOpt、SkillRL和协同进化技能（CoEvoSkills）提出不同的修订或学习循环（Liu 等人，2026b；Li 和 Hu，2026；Yang 等人，2026；Xia 等人，2026；Zhang 等人，2026b）。最近的实证工作扩展了研究对象：从原始经验到技能消费（Skill Consumption）将经验生成、提取和消费分开（Huang 等人，2026）；智能体技能在现实环境中的表现如何（How Well Do Agentic Skills Work in the Wild）研究现实中的技能检索（Liu 等人，2026a）；SkillCoach评估流程级技能使用（Zhu 等人，2026）；以及SkillJuror隔离技能组织（skill organization）的影响（Chen等人，2026）。

realistic skill retrieval (Liu et al. 2026a); SkillCoach evaluates process-level skill use (Zhu et al. 2026); and SkillJuror isolates the effect of skill organization (Chen et al. 2026).

These studies show that outcome-only evaluation is already recognized as incomplete. Our focus is narrower and complementary: when a persistent skill is revised and the post-revision score rises, what evidence identifies a retained artifact effect rather than a selected or measurement-dependent endpoint? We therefore organize evidence by lifecycle contrast rather than by method score.

Evaluation and Attribution of Agent Improvement

Agent evaluation increasingly examines process quality, sequential adaptation, verifier reliability, and counterfactual behavior. Counterfactual Trace Auditing measures how a provided skill changes an agent trajectory (Zhou et al. 2026a). SEA-Eval evaluates self-evolving agents beyond isolated episodes (Jiang et al. 2026). Co-evolving verifier work and grader audits study interactions between the policy and its measurement signal (Zhang et al. 2026b,c). More broadly, reward hacking and test-set contamination research demonstrates that performance on a proxy or exposed test need not imply the intended capability (Amodei et al. 2016; Oren et al. 2023).

We combine these concerns within the lifecycle of persistent skill revision. The central distinction is between retention, behavioral scope, evolution-and-selection pathway, and output-level measurement dependence. Information-integrity checks play a supporting role: they bound direct answer or oracle exposure, but do not convert every verifier-aware adaptation into a leakage claim.

Limitations

Our evidence is mechanism-focused rather than a benchmark-wide prevalence study. The 24-task retention panel is purposive and contains only two fresh deployments per condition; it exposes instability but cannot estimate deployment rates. The controlled Invoice, Energy, SearchQA, Radar, Spreadsheet, and OfficeQA probes use small, heterogeneous task families rather than a fully crossed method-by-intervention design. SkillRevise, for example, lacks matched artifact surgery and output-locked measurement evidence of the same strength, while the strongest verifier swap is concentrated in the Radar family. We therefore do not claim that any observed mechanism is universal or that one method outperforms another.

Several conclusions are also bounded by observability. Output-locked comparisons identify measurement dependence but do not determine which verifier best captures intended behavior without an external specification. Artifact forensics and wrong-metadata controls can rule out simple direct exposure, but not implicit answer-following policies or every form of contamination. Finally, our scope is frozen-agent, feedback-driven systems with persistent external artifacts; parameter-updating self-evolution, multimodal skills, and online environments may exhibit different lifecycle dynamics. Confirmatory evidence will require a shared sampling plan that fully crosses actual parent and final artifacts, fresh repeats, controlled variants, and calibrated verifiers.

Conclusion

We study whether benchmark gains from self-evolving persistent skills correspond to retained behavioral improvement and, independently, which lifecycle mechanisms produce those gains. In the recorded variant evaluations, SkillEvolver retains positive mean gains across changes in input representation, task-instance structure, and coupled requirements; SPARK shows gains on two types of change, while CoEvoSkills shows declines on all three. Execution traces connect successful cases to reusable diagnostic steps and preservation of task constraints. Our audits show that revised skill artifacts can produce real behavioral changes, but selected gains can disappear under fresh deployment and the transfer established so far remains narrow. Feedback, optimizer variation, validation selection, artifact content, execution variability, and verifier choice can each shape the observed endpoint.

A higher benchmark score should therefore be reported only at the strongest level supported by matched evidence: a selected gain, a retained artifact effect, bounded controlled transfer, evolution-pathway dependence, or measurement dependence. We provide a lifecycle audit that links each claim to the required intervention and reporting evidence. Claims of transferable skill learning require frozen parent-to-final improvement that survives fresh deployment and predeclared task variation under valid measurement; weaker evidence should be reported more narrowly.

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SkillCoach评估流程级技能使用（Zhu 等人，2026）；以及SkillJuror隔离技能组织（Chen等人，2026）。

这些研究表明，仅基于结果的评估已被公认为是不完整的。我们的关注点更狭窄且具有补充性：当持久的技能被修订且修订后的得分上升时，什么样的证据能够识别出保留的工件效应，而不是被选择或测量依赖的终点？因此，我们按生命周期对比来组织证据，而不是按方法得分来组织。

代理改进的评估与归因

代理评估越来越多地考察过程质量、顺序适应、验证器可靠性和反事实行为。反事实轨迹审计测量提供的技能如何改变代理轨迹（Zhou 等人，2026a）。SEA-Eval 超越孤立场景评估自进化代理（Jiang 等人，2026）。协同进化的验证器工作和评分器审计研究政策与其测量信号之间的交互（Zhang 等人，2026b,c）。更广泛地说，奖励破解和测试集污染研究证明，在代理或暴露的测试上的表现不一定意味着预期的能力（Amodei 等人，2016；Oren 等人，2023）。

我们将这些关注点结合在持续技能修订的生命周期内。核心区别在于保持性、行为范围、进化与选择路径以及输出级测量依赖性。信息完整性检查起辅助作用：它们限制直接答案或预言机暴露，但不会将所有验证器感知的适应性都转化为泄漏声明。

局限性

我们的证据是机制导向的，而非跨基准的普遍性研究。24任务保持性面板是经过设计的，每个条件下仅包含两次全新部署；它暴露了不稳定性，但无法估计部署率。受控的Invoice、Energy、SearchQA、Radar、Spreadsheet和OfficeQA探测使用小型异构任务族，而非完全交叉的方法-干预设计。例如，SkillRevise缺乏匹配的工件手术和输出锁定测量证据，而最强的验证器交换集中在Radar族。因此，我们不声称任何观察到的机制是普遍的，也不声称一种方法优于另一种方法。

一些结论也受限于可观察性。输出锁定的比较可以识别测量依赖性，但在没有外部规范的情况下无法确定哪个验证器最能捕捉预期行为。工具体验和错误元数据控制可以排除简单的直接暴露，但无法排除隐式答案跟随策略或任何形式的污染。最后，我们的范围是冻结代理、反馈驱动的系统以及持久的工具体现；参数更新自进化、多模态技能和在线环境可能表现出不同的生命周期动态。确认性证据将需要一个共享的采样计划，该计划完全跨越实际父代和最终工具体现、全新重复、受控变体和校准验证器。

结论

我们研究自进化持久技能的基准收益是否与保留的行为改进相对应，并且独立地，哪些生命周期机制产生了这些收益。在记录的变体评估中，SkillE-volver在输入表示、任务实例结构和耦合需求的变化中始终保持正向平均收益；SPARK在两种类型的变化中显示收益，而CoEvoSkills在三种变化中均显示下降。执行轨迹将成功案例与可重用的诊断步骤和任务约束的保留联系起来。我们的审计表明，修订的技能工具体现可以产生真实的行为变化，但在全新部署下，选定的收益可能会消失，迄今为止建立的迁移范围仍然狭窄。反馈、优化器变体、验证选择、工具体现内容、执行变异性以及验证器选择都可以塑造观察到的终点。

因此，只有在有匹配证据支持的最强级别时，才应报告更高的基准分数：一个选定的收益、一个保留的工件效应、有界的受控迁移、进化路径依赖性或测量依赖性。我们提供生命周期审计，将每个主张与所需的干预措施和报告证据联系起来。可迁移技能学习的主张需要冻结的从父级到最终改进，该改进在全新部署和预声明的任务变化下在有效测量下仍然存在；较弱的证据应更严格地报告。

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